RIVR Health.AI – Prototype Requirements

Feature Set: Health Record Ingestion, Timeline, ER 3×5 Card, Annual Checkup Summary, QR Sharing, Wearable Ingestion

1. Product Goal (Prototype)

Build an MVP that lets a user:

1. Upload or scan existing medical records (PDFs / images).
2. Extract core clinical information into a structured profile and event timeline.
3. Auto-generate:
   - A concise “ER 3×5 Card” summary for EMTs / ER physicians.
   - A written “Annual Checkup Summary” with key history and recent updates.
   - Create real time Patient Health timeline
4. Share this information securely via a QR code.
5. Ingest biometric data from wearables into the same user profile (even if the first version is minimal).

All functionality is tied to a **user account**.

2. Core Data Concepts

2.1 User Profile

Minimum fields:

- UserID
- First name, last name
- Date of birth
- Sex / Gender
- Occupation
- Marital status
- Number of children (optional)
- Contact info (email, mobile)
- Emergency contact (optional)

2.2 Clinical “History” Buckets

These will be populated from OCR + structured inputs:

- **Social History (SocHx)**
- **Surgical History (PSHx)**
- **Family History (FamHx)**
- **Past Medical History (PMHx)**
- **Allergies**
- **Medications**
- **Hospitalizations**
- **Vitals**
  - Height / Ht
  - Weight / Wt
  - BMI
  - Pulse
  - (room for expansion: BP, temp, etc.)

2.3 Events / Timeline Items

Every medical event should be represented as a structured **Event** with:

- EventID
- UserID
- **Date (required)** – if unavailable, allow an approximate date or "year only".
- Event type (e.g., "Annual Checkup", "Surgery", "Hospitalization", "Clinic Visit", "Lab Result", "Imaging", "Symptom Episode")
- Source (OCR'd document, manual entry, wearable, etc.)
- Free-text description
- Linked data:
  - Updates to PMHx, PSHx, Meds, Allergies, Vitals, etc.
  - Files/attachments (original PDF, images)

The timeline is a chronological list of Events, updated whenever new data is ingested.

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3. OCR Ingestion & Data Extraction

3.1 Inputs

- User uploads:
  - PDFs (clinic notes, discharge summaries, lab reports, etc.).
  - Images (photos of documents) → processed via OCR to text.

3.2 OCR Pipeline

1. **Upload** file(s) mapped to authenticated UserID.
2. Run OCR (service of choice) to extract raw text.
3. Store:

    - Original file
    - OCR raw text
    - Metadata (upload date, file name, source)

3.3 Subheading Extraction

From the OCR'd text, automatically identify and parse subheadings into structured data:

Target headings / sections:

- Social History
- Surgical History
- Family History
- Past Medical History / PMHx
- Allergies
- Medications
- Hospitalizations
- Vitals (Height, Weight, BMI, Pulse)

Requirements:

- Recognize common variants and abbreviations (e.g., "SocHx", "PMH", "Past Medical Hx", "Meds", "Allergies/Adverse Reactions", etc.).
- For each heading, capture:
    - Heading type (mapped to one of the standard buckets above)
    - Content (the text under that heading until the next heading or end of section)
    - Any dates or temporal information associated with items (e.g., surgery dates, hospitalization dates).

3.4 Event Creation

- For each document, create at least one **Event** (e.g., "Clinic Visit – [Date if available]").
- Within the event, map extracted data into:
    - PMHx, PSHx, Meds, FamHx, SocHx, Allergies, Hospitalizations, Vitals, etc.
- When explicit dates are found for specific items (e.g., "Right shoulder labrum surgery – 2015"), create or update separate **sub-events** or attributes with that date.

If no date is found in the document, prompt the user in the UI to add/confirm an approximate date (for the prototype this can be a simple manual edit step).

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4. 3×5 Digital ER Card Summary

The 3×5 card is a **single screen, short summary** designed to be quickly read by EMTs / ER docs. It should pull from structured data, not raw text.

4.1 Data Used

From profile + structured history:

- Age (computed from DOB)
- Sex / Gender
- Occupation
- Marital status
- Children (optional short phrase)
- Health status / activity level (from SocHx / user input)
- **PMHx** (top 3–5 most relevant conditions)
- **PSHx** (top surgeries)
- **Smoking status** (Smoker / Non-Smoker)
- **Alcohol use** (Social ETOH drinker / frequency / quantity)
- **Allergies** (NKDA or specific allergy list, e.g., "Allergic to PCN")
- **Current medications** (list or truncated list with "+X more" if long)
- Recent key symptoms for current episode (optional free text input)

4.2 Sentence Template

The system should synthesize a concise, human-readable paragraph like:

```
51yo Male, Financial Advisor, in relatively good health, married for 15 years
with 3 children (10, 6, 3), runs 3x/week and weight trains 2x/week, with PMHx
of HTN (hypertension), hypercholesterolemia, no PSHx, non-smoker, social ETOH
drinker on weekends (~5–6 beers), NKDA, currently taking Norvasc 5mg once daily
and managing cholesterol with diet and exercise.
```

Optionally followed by a short line describing the **current issue** (if entered):

```
Presenting with headache x5 days after 10 days of cold symptoms, taking extra
Tylenol, dizziness, and two episodes of vomiting in last 24 hours.
```

4.3 UX Requirements

- Card should fit comfortably on a mobile screen and printable "3×5" size.
- Design as a simple component:
  - Title: "Emergency Summary – [First Name Last Initial]"
  - Body: 1–2 short paragraphs as above.
- Allow export as:
  - Screen view
  - PDF or image (for future)
  - Accessible via QR code (see below).

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5. Annual Checkup Summary (with Dates)

Generate a structured text block summarizing the user's health status around an annual checkup.

5.1 Sections

Using structured data and timeline events, auto-populate:

- **PMHx:** list of conditions (with key dates/diagnosis years)
- **PSHx:** list of surgeries with dates
- **Meds:** current medication list (name, dose, frequency)
- **Supplements:** (if captured)
- **All:** allergies (NKDA vs. list)
- **SocHx:** smoking status, alcohol use, exercise, occupation, housing, etc.
- **FamHx:** major family conditions (e.g., CAD, cancer, diabetes)

5.2 New Updates

Below the core history, generate sections:

- **New Updates of Health / Visits to Other Providers / Diagnoses:**
  Auto-pull any events since the last annual checkup date and list them by date and type.
- **Last Medical Providers Visits / Dates:**
  List recent visits: date, provider name (if available), visit type.
- **Labs:**
  Summaries of recent key labs (e.g., A1C, lipid panel) if available.
- **Images (Imaging):**
  Recent imaging events (X-ray, MRI, CT) with dates and short descriptions.
- **Updated Vitals from Other Providers / Home:**
  Latest vitals (BP, HR, weight, BMI, etc.) from events or wearable data.

This should render as a structured, readable summary a clinician could skim quickly.

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6. Personal Health Timeline

Create a dynamic, chronological **timeline view** of the user's life events:

- Each **Event** (see section 2.3) appears with:
  - Date
  - Title (e.g., "Annual Checkup – PCP", "Right Shoulder Surgery", "Hospitalization – Pneumonia")
  - Key tags: PMHx / PSHx / Meds / Labs / Imaging / Vitals / Symptom
- The timeline automatically updates whenever:
  - New records are OCR'd and parsed.
  - User edits/adds events.

    - Wearables send new data (for now, may be aggregated into "Vitals / Activity" entries, daily or weekly).

This is the "living chart" that the ER card and annual summaries are derived from.

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7. QR Code Sharing

The user needs a **QR code** that an EMT or doctor can scan to instantly access the ER 3×5 Card and optionally the broader summary.

Requirements:

- QR encodes a **short URL** to our backend, not raw PHI.
- URL resolves to:
    - A read-only view of the ER 3×5 Card.
    - Optionally a "View More Details" link (Annual Checkup Summary / timeline snapshot) – can be a phase-2 extension.
- Must be **access-controlled**:
    - For MVP, acceptable implementation: shareable link with a random, high-entropy token.
    - Later, can add PIN or time-limited access.

UX:

- In user profile, show "Emergency QR Code" view.
- Allow user to download/save QR image to phone (e.g., wallet, photos).

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8. Wearable / Biometric Data Ingestion (Prototype Scope)

Goal: Make it straightforward to include biometrics in the user's timeline and vitals.

MVP requirements:

- **Data model** for biometric readings:
    - UserID
    - Source (e.g., "Apple Watch", "Fitbit", "Manual entry")
    - Measurement type (HR, steps, sleep, SpO2, BP, weight, etc.)
    - Value
    - Unit
    - Timestamp
- **API / ingestion endpoint** that can accept batches of biometric readings (JSON).
- UI:

    - Basic display of recent vitals (e.g., last HR, resting HR trend, last weight).
    - These readings can populate or update "Vitals" sections for:
        - Timeline
        - Annual Checkup Summary.

Actual integrations (Apple Health, Google Fit, etc.) can be simulated with manual uploads or a simple mock connector for the prototype.

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9. Insightful Questions (Onboarding Questionnaire) ENRICH Profile

During onboarding (or in a separate "Story" section), collect qualitative data to better understand the patient's context.

Questions to present (free-text answers, all optional but encouraged):

1. Tell me about your relationships that are the most important to you and why.
2. Tell me how you would describe your health approach. What does "being healthy" look like to you?
3. Tell me about a positive memory you have from childhood. How old were you?
4. Tell me about your parents' relationship when you were growing up. How did you feel with them and your siblings?
5. What are things you are good at in terms of health, and what are things that are difficult for you?
6. How would you describe the season of life you're in right now?
7. What roles feel most important to you right now (parent, partner, worker, caregiver, etc.)?
8. On a typical day, what takes most of your time and energy?
9. If you had an extra free hour most days, how would you honestly want to use it?
10. When you hear the word "health," what comes to mind first?

Store answers as part of the user profile (e.g., `NarrativeAnswers[]`) so they can be referenced in future coaching/UX flows but **not** shown on the ER 3×5 card.
